

ALFAA77085

## Gallium (III) nitrate, hydrate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | Gallium (III) nitrate, hydrate<br>Gallium (III) nitrate, hydrate                                                                                                                                                                                                   |
| Cat No. :                     | 77085                                                                                                                                                                                                                                                              |
| Synonyms                      | Gallium nitrate; Nitric acid, gallium (3+) salt; Gallium trinitrate                                                                                                                                                                                                |
| CAS No                        | 69365-72-6                                                                                                                                                                                                                                                         |
| Molecular Formula             | Ga N3 O9 . x H2 O                                                                                                                                                                                                                                                  |
| Supplier                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                   |                     |                  |
|-------------------------------------------------------------------------------------------------------------------|---------------------|------------------|
| Physical State<br>Powder Solid                                                                                    | Appearance<br>White | Odor<br>Odorless |
| <b>Emergency Overview</b><br>May intensify fire; oxidizer. Causes skin irritation. Causes serious eye irritation. |                     |                  |

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Oxidizing solids                  | Category 2 |
| Skin Corrosion/Irritation         | Category 2 |
| Serious Eye Damage/Eye Irritation | Category 2 |

#### Label Elements



Signal Word

Danger

## Gallium (III) nitrate, hydrate

**Hazard Statements**

H272 - May intensify fire; oxidizer  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P220 - Keep away from clothing and other combustible materials  
P221 - Take any precaution to avoid mixing with combustibles  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P337 + P313 - If eye irritation persists: Get medical advice/attention  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Oxidizing. Contact with combustible material may cause fire.

**Health Hazards**

Causes skin irritation. Causes serious eye irritation.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                | CAS No     | Weight % |
|--------------------------|------------|----------|
| Gallium nitrate, hydrate | 69365-72-6 | 100      |
| Gallium(III) nitrate     | 13494-90-1 | -        |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.

**Ingestion**

Do NOT induce vomiting. Get medical attention.

**Most important symptoms and effects**

No information available.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of

## Gallium (III) nitrate, hydrate

contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Prevent further leakage or spillage if safe to do so.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Ensure adequate ventilation. Wear personal protective equipment/face protection. Keep away from clothing and other combustible materials. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Minimize dust generation and accumulation. Wash hands before breaks and immediately after handling the product.

**Storage**

Keep in a dry, cool and well-ventilated place. Do not store near combustible materials. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and

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inhalable dust

## Exposure Controls

## Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ventilation systems. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

**Appearance** White  
**Physical State** Powder Solid

**Odor** Odorless  
**Odor Threshold** No data available  
**pH** No information available  
**Melting Point/Range** 110 °C / 230 °F  
**Softening Point** No data available  
**Boiling Point/Range** No information available  
**Flash Point** No information available

**Method -** No information available

**SAFETY DATA SHEET****Gallium (III) nitrate, hydrate**

|                                                |                          |       |
|------------------------------------------------|--------------------------|-------|
| <b>Evaporation Rate</b>                        | Not applicable           | Solid |
| <b>Flammability (solid,gas)</b>                | No information available |       |
| <b>Explosion Limits</b>                        | No data available        |       |
| <b>Vapor Pressure</b>                          | No information available |       |
| <b>Vapor Density</b>                           | Not applicable           | Solid |
| <b>Specific Gravity / Density</b>              | No data available        |       |
| <b>Bulk Density</b>                            | No data available        |       |
| <b>Water Solubility</b>                        | No information available |       |
| <b>Solubility in other solvents</b>            | No information available |       |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |       |
| <b>Autoignition Temperature</b>                | Not applicable           |       |
| <b>Decomposition Temperature</b>               | No data available        |       |
| <b>Viscosity</b>                               | Not applicable           | Solid |
| <b>Explosive Properties</b>                    | No information available |       |
| <b>Oxidizing Properties</b>                    | Oxidizer                 |       |
| <b>Molecular Formula</b>                       | Ga N3 O9 . x H2 O        |       |
| <b>Molecular Weight</b>                        | 255.74                   |       |

**SECTION 10. STABILITY AND REACTIVITY**

|                                 |                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions. Oxidizer: Contact with combustible/organic material may cause fire. |
| <b>Hazardous Reactions</b>      | No information available.                                                                           |
| <b>Hazardous Polymerization</b> | No information available.                                                                           |
| <b>Conditions to Avoid</b>      | Avoid dust formation. Incompatible products. Excess heat. Combustible material.                     |
| <b>Materials to avoid</b>       | Strong reducing agents. Combustible material. Oxidizing agent.                                      |

**Hazardous Decomposition Products** Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 11. TOXICOLOGICAL INFORMATION**

|                                               |                                                             |
|-----------------------------------------------|-------------------------------------------------------------|
| <b>Product Information</b>                    | No acute toxicity information is available for this product |
| <b>(a) acute toxicity;</b>                    |                                                             |
| <b>(b) skin corrosion/irritation;</b>         | Category 2                                                  |
| <b>(c) serious eye damage/irritation;</b>     | Category 2                                                  |
| <b>(d) respiratory or skin sensitization;</b> |                                                             |
| <b>Respiratory</b>                            | No data available                                           |
| <b>Skin</b>                                   | No data available                                           |
| <b>(e) germ cell mutagenicity;</b>            | No data available                                           |
| <b>(f) carcinogenicity;</b>                   | No data available                                           |
|                                               | There are no known carcinogenic chemicals in this product   |
| <b>(g) reproductive toxicity;</b>             | No data available                                           |

## Gallium (III) nitrate, hydrate

|                                            |                           |
|--------------------------------------------|---------------------------|
| (h) STOT-single exposure;                  | No data available         |
| (i) STOT-repeated exposure;                | No data available         |
| Target Organs                              | No information available. |
| (j) aspiration hazard;                     | Not applicable<br>Solid   |
| Symptoms / effects, both acute and delayed | No information available  |

## SECTION 12. ECOLOGICAL INFORMATION

|                                                                                              |                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects                                                                          | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.                                                                                   |
| Persistence and Degradability<br>Degradability                                               | No information available<br>Not relevant for inorganic substances.                                                                                                                                            |
| Bioaccumulative Potential                                                                    | No information available                                                                                                                                                                                      |
| Mobility in soil                                                                             | No information available                                                                                                                                                                                      |
| Endocrine Disruptor Information<br>Persistent Organic Pollutant<br>Ozone Depletion Potential | This product does not contain any known or suspected endocrine disruptors<br>This product does not contain any known or suspected substance<br>This product does not contain any known or suspected substance |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                     |                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point.                                                                                              |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                                          |

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

|                         |                             |
|-------------------------|-----------------------------|
| UN-No                   | UN1477                      |
| Proper Shipping Name    | NITRATES, INORGANIC, N.O.S. |
| Technical Shipping Name | Gallium nitrate hydrate     |
| Hazard Class            | 5.1                         |
| Packing Group           | II                          |

IMDG/IMO

## SAFETY DATA SHEET

## Gallium (III) nitrate, hydrate

UN-No UN1477  
Proper Shipping Name NITRATES, INORGANIC, N.O.S.  
Technical Shipping Name Gallium nitrate hydrate  
Hazard Class 5.1  
Packing Group II

IATA

UN-No UN1477  
Proper Shipping Name NITRATES, INORGANIC, N.O.S.  
Technical Shipping Name Gallium nitrate hydrate  
Hazard Class 5.1  
Packing Group II

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|--------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Gallium nitrate, hydrate | -                                                   | -                                       | X    | -     | -         | -    | -   | -     | -    |      | -    | -        |
| Gallium(III) nitrate     | X                                                   | -                                       | X    | X     | 236-815-5 | X    | -   | -     | -    | X    | X    | KE-17446 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
Creation Date 26-Sep-2009  
Revision Date 24-Apr-2024  
Revision Summary New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**DNEL** - Derived No Effect Level  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**PNEC** - Predicted No Effect Concentration  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**